

Chemical Shift Encoding-Based Water-Fat Separation

14

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14.1 Introduction

Chemical shift encoding-based imaging (CSI) tries to differentiate chemical species based on their chemical shift property that originates from the shielding and deshielding effects caused by the electron density around the nucleus. In the context of proton-based water–fat imaging (WFI), this means that most protons of the fat component experience higher shielding and thus precess at lower frequencies. The frequency is measured in ppm units as a fraction of the absolute resonance frequency. In WFI at body temperature, the reference frequency of water is at about 4.67 ppm, while the reference frequency of the methylene peak (the main fat peak) is at about 1.30 ppm. This difference in chemical shift that corresponds to a frequency difference depending on the actual experienced field strength is encoded in the echo time dimension. For example, at 3T, the main frequencies of water and fat are 430 Hz (3.37 ppm) apart, corresponding to a phase cycling period of approximately 2.3 ms. Consequently, echoes acquired at TEs of $n \cdot 2.3$ ms are called in-phase and at TEs of $(n \cdot 2.3 + 1.15)$ ms are called out-of-phase. The signal can then be acquired at different echo times and decomposed into its water and fat signal using an appropriate model (Fig. 14.1).

The idea of CSI was first introduced in a publication by eponymous W. Thomas Dixon in 1984 [1], and, since then, the use of WFI has significantly grown over the years both in clinical diagnostic imaging and in the research setting. WFI enables efficient fat suppression especially in experimental settings and anatomical locations where significant transmit B_1 and B_0 inhomogeneities can be observed. WFI can thus overcome some of the challenges faced by traditional chemical selective-fat suppression and inversion-based fat suppression [2]. In addition, WFI enables the simultaneous acquisition of coregistered nonfat-suppressed and fat-suppressed images in a single scan that can be beneficial in diagnostic imaging (albeit at the cost of prolonged scan time). Finally, WFI has recently gained considerable attention by enabling the quantification of the proton density fat fraction (PDFF), which constitutes a standardized metric of tissue fat concentration [3]. PDFF mapping is currently used in multiple organs (liver, pancreas, skeletal muscle, bone marrow) with a wide range of applications in fatty liver disease, metabolic disorders (obesity, diabetes), and musculoskeletal diseases [4].

WFI requires estimating the water and fat signal components in the presence of field-map variations. In addition, if non-Cartesian imaging is employed or quantitative parameter maps are of interest, this simple concept often requires to be extended to take additional effects into account and correct for

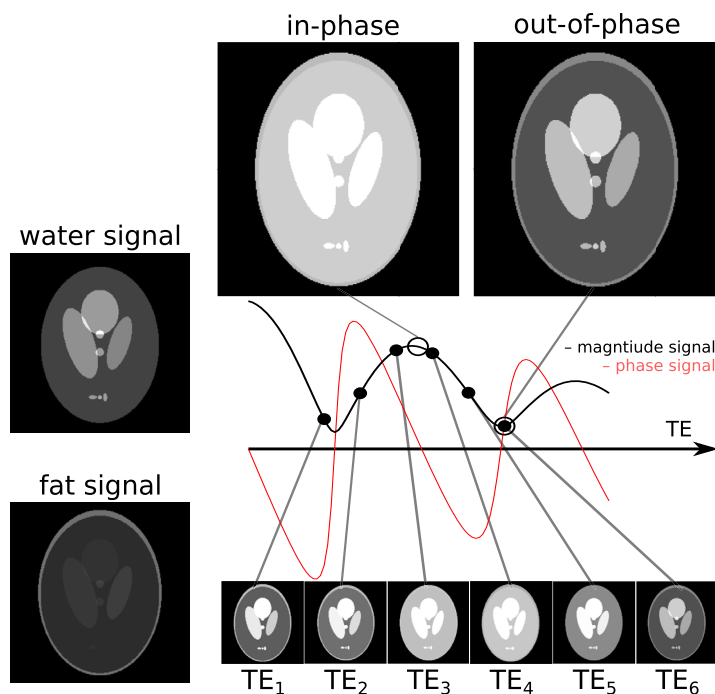


FIGURE 14.1 Schematic concept of chemical shift encoding-based signal encoding

On the left: The column on the left shows the ground truth water (top) and fat (bottom) signal of a simple numerical water-fat phantom. Right top row depicts characteristics contrasts for in-phase and out-of-phase images of the phantom that can be acquired at distinct echo times. Right middle row: A conceptional signal evolution for a water-fat signal (magnitude and phase-signal components) with a proton density fat fraction (PDFF) of 50%. Right bottom row: Example contrast of the same phantom sampled at six different echo times with equidistant echo time spacing. In order to separate the water from the fat signal, multiple sampling points (traditionally six points) are used to fit a water-fat model to the signal.

various sources of errors. Therefore, WFI requires multiple considerations in the data acquisition and signal modeling, which essentially include the whole chain of image reconstruction. In the following, the most important aspects in the reconstruction of WFI are presented.

14.2 Theory on chemical species separation

We first explain the chemical shift property, discuss the chemical shift property of fat, and then introduce the general formulation for chemical species separation.

14.2.1 The chemical shift property

In many quantitative MRI parameter-estimation techniques, the time evolution of the net magnetization of an ensemble of protons in a gradient echo or a free induction decay signal is the central focus. In human tissue these protons are always bound in molecules and therefore surrounded by an electron cloud defining the protons' chemical environment. Electron clouds are subject to polarization by the main magnetic field and therefore generate their own magnetic field based on the magnetic susceptibility of the molecule. The polarization of electrons during an MRI experiment, among others, gives rise to the effect of chemical shift affecting the MR signal dynamics [5].

For observer protons inside a molecule, the field created by the polarized electron cloud is known as the demagnetization field [6], which is dependent on the chemical surrounding and the local geometry of the proton's location inside the molecule (electron configuration, angle and length of chemical bonds between nuclei, etc.). According to Lenz's law [7], the demagnetization field is opposed to the polarizing field, the main magnetic field B_0 in MRI. Protons in a specific chemical environment, a so-called chemical species denoted by subscript p , therefore experience a reduced field strength and precess with a specific Larmor precession frequency defined by their shielding constant δ_p

$$\omega_p = -\gamma B_0(1 - \sigma_p) = \omega_0(1 + \delta_p). \quad (14.1)$$

By convention the chemical shift in MRI (and magnetic resonance spectroscopy) is often given in units of the deshielding constant δ_p [8]. Compared to water, most molecules in human tissue have a higher electron density and therefore a negative deshielding constant with respect to the assumed center frequency ω_0 tuned to the protons in water molecules. In the MR signal evolution, the presence of a single chemical species denoted p adds an additional magnetic-field term to the precession of an observer proton

$$S(t) \sim \int_V m(\mathbf{r}, t) \exp(-i\omega_0(1 + \delta_p)t) d\mathbf{r}. \quad (14.2)$$

The factor $\omega_0\delta_p$ leads to a modulation of the periodic precession of the magnetization around the main magnetic field with frequency ω_0 . Therefore, the dynamics of the signal $S(t)$ over multiple echoes effectively encodes the presence of species p in the modulated frequency of the signal phase.

14.2.2 The chemical shift of fat

In body MRI, the most relevant molecules besides water are lipids due to their high abundance and chemical shifts properties. The lipid signal has a more complex spectral appearance compared to the water signal, which is usually well approximated with a single resonance frequency (Fig. 14.2) In MRI, all fat signals are typically assumed to arise exclusively from triglycerides due to their dominant abundance. Thus, the spectral model of the fat signal is based on the chemical structure of triglycerides. Triglycerides hold a number of different chemical species, each with their own specific chemical shifts. The most abundant chemical species in a triglyceride is the methylene group with its specific chemical shift of $\delta_{CH_2} = 1.3$ ppm.

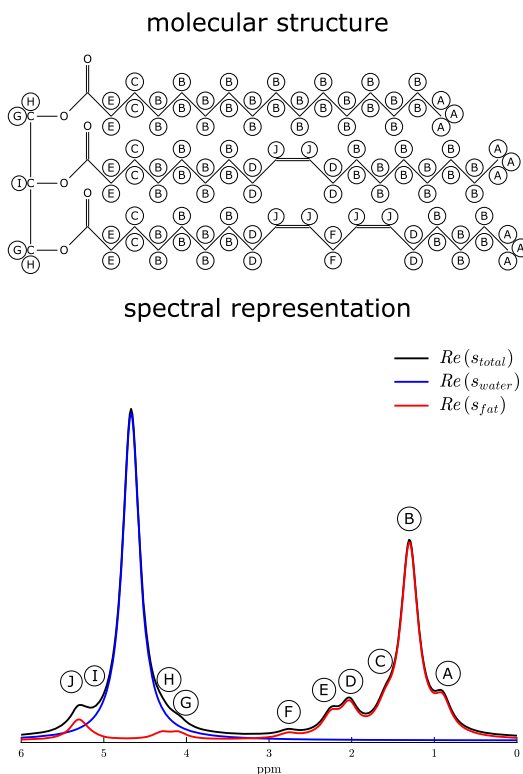


FIGURE 14.2 Chemical structure of a triglyceride and corresponding ^1H proton spectral appearance

The chemical structure of a triglyceride is characterized by a glycerol (on the left side) esterified with three fatty acids (palmitic acid, oleic acid, and linoleic acid from top to bottom on the right side). The circled letters (A–J) label and group the protons according to their specific resonance frequency. The corresponding spectral appearance of a mixture (black line) of triglycerides (red (mid gray in print version) line) and water (blue (dark gray in print version) line, peak at 4.67 ppm) is displayed below and shows the characteristic peaks that can be assigned to a corresponding proton environment in the chemical structure (see labels A–J).

14.2.3 Signal model for water–fat separation

Chemical shift encoding-based imaging uses acquisitions of multiple echoes at increasing echo times to estimate the contributions of possibly several chemical species to the MR signal, yielding important biomarkers like, e.g., the proton density fat fraction and other specific parameters like R_2^* .

The time evolution of the complex MR signal inside a single voxel, $s_n \equiv s(t_n)$, can be written in general as

$$s_n = \sum_{p=1}^P \varrho_p e^{i\phi_p} e^{(i\omega_p - r_p)t_n}, \quad (14.3)$$

where the contribution to the signal from chemical species, $p = 1, \dots, P$, is characterized by their magnitude ϱ_p , phase after the RF-excitation (at $t = 0$) ϕ_p , resonance frequency ω_p , and transverse relaxation rate r_p .

Since the estimation of the whole set of model parameters $\{\varrho_p, \phi_p, \omega_p, r_p\}$, $p = 1, \dots, P$ in Eq. (14.3) would require an impractical amount of echoes s_n , it is necessary to reduce the number of parameters on the right-hand side of Eq. (14.3). A typical example in WFI is the reduction of Eq. (14.3) to the widely used multipeak single- R_2^* model by fixing the chemical shifts of all $p \in 2, \dots, 10$ fat peaks—assuming the first peak $p = 1$ corresponds to the water peak—to a priori values normally obtained from MR spectroscopy by setting $\omega_m = \omega_1 + \delta\omega_p$, $p \in 2, \dots, P$. Furthermore, all peaks, including water, are assumed to decay with the same relaxation time R_2^* , which finally yields

$$\hat{s}_n = (W + c_n F) e^{(i2\pi f_B - r_2^*)t_n}, \quad c_n = \sum_{p=1}^P a_p e^{i2\pi \delta f_p t_n}, \quad \text{with} \quad \sum_{p=1}^P a_p = 1, \quad (14.4)$$

where we substituted $W = \varrho_1 \exp(i\phi_1)$ and $f_B = \omega_1/2\pi$ for the water peak and $F\alpha_p = \varrho_p \exp(i\phi_p)$, $p \in \{1, \dots, P\}$ for the fat peaks, introducing f_B for the field-map and α_p for the weight of the p -th fat peak in the fat spectrum model.

The signal model given by Eq. (14.4) describes the signal in a single observation at acquired echo time t_n . The definition of the following model matrix [9,10]:

$$\mathbf{A}(f_B, R_2^*) = \text{diag} \left[e^{(i2\pi f_B - R_2^*)t_1}, \dots, e^{(i2\pi f_B - R_2^*)t_n} \right] \begin{pmatrix} 1 & c_1 \\ \vdots & \vdots \\ 1 & c_N \end{pmatrix}, \quad (14.5)$$

enables us to write the signal model in a multi-observation matrix formulation

$$\hat{\mathbf{s}} = [s_1, \dots, s_n]^T = \mathbf{A}(f_B, R_2^*)[W, F]^T.$$

14.3 Solving the water–fat separation problem

14.3.1 Parameter estimation in water–fat separation

The least-squares parameter estimation problem can then be stated as

$$W, F, f_B, R_2^* = \underset{W', F', f_B', R_2^{*'}}{\operatorname{argmin}} \left\| \mathbf{s} - \hat{\mathbf{s}}(W', F', f_B', R_2^{*'}) \right\|_2^2. \quad (14.6)$$

By means of the VARPRO [11], the linear parameters in Eq. (14.6) can be substituted by

$$[W, F]^T = \mathbf{A}^+(f_B, R_2^*)\hat{\mathbf{s}}, \quad (14.7)$$

where $\mathbf{A}^+ = \mathbf{A}^\dagger(\mathbf{A}^\dagger\mathbf{A})^{-1}$ is the Moore–Penrose pseudo-inverse of matrix in Eq. (14.5), resulting in the optimization

$$f_B, R_2^* = \underset{f'_B, R_2^{*'}}{\operatorname{argmin}} \left\| (1 - \mathbf{A}\mathbf{A}^+) \hat{\mathbf{s}} \right\|_2^2, \quad (14.8)$$

which only minimizes the nonlinear parameters f_B, R_2^* in the signal model of Eq. (14.4) since the linear parameters W and F are determined by Eq. (14.7).

The variable projection (VARPRO) minimization problem of Eq. (14.8) is specific to the single- R_2^* water–fat signal model of Eq. (14.4), however it can be generalized to the broader class of weighted sums of complex exponentials with phase terms linearly varying in the echo time [12].

The minimization in Eq. (14.8) can in general be solved by standard nonlinear fitting but requires two additional considerations: the field map estimation and the analysis of the noise performance of the estimate parameters.

14.3.2 The field-map estimation problem

The field-map parameter f_B describes local frequency shifts due to static field inhomogeneity. The static field inhomogeneity mainly originates from the large magnetic susceptibility difference between air and tissue and the geometry of the subject placed inside the scanner. Additionally, different magnetic susceptibility of different tissues contributes to the field inhomogeneity.

The field-map parameter f_B is subject to phase wrapping, and its value can only be determined up to modulo 2π . Traditionally, WFI primarily focuses on the extraction of the water and the fat-signal parameters, where a wrapped field-map is not of importance. However, in cases where further processing of the field-map is needed, the wrapping needs to be taken into account. In addition, and also connected to the complex exponential with the time-dependent argument scaled by the field-map parameter, the least-squares cost function in Eq. (14.8) is nonconvex in the field-map parameter.

The nonlinearity of the field-map parameter often leads to the occurrence of multiple local minima of the cost function that can correspond to unphysical parameter combinations. A convergence of the field-map to such local minima can lead to the infamous water–fat swaps, where the water and the fat signals in a voxel are incorrectly assigned [13]. Water–fat swaps can in particular occur in regions with a rapid spatial variation of the field-map and/or low signal to noise ratio, where the number of local minima of the cost function increases. Therefore, iterative nonlinear optimization schemes require an initialization of the field-map close to the true parameter value to ensure convergence to the global minimum. Finding a good initialization of the field-map often necessitates the incorporation of additional information to the single-voxel minimization problem (14.8). Common approaches involve the incorporation of prior knowledge from several field-map contributions to the problem [14,15] or the incorporation of neighborhood information of each voxel by setting up a global cost function for all voxels similar to

$$f_B, R_2^* = \underset{f'_B, R_2^{*'}}{\operatorname{argmin}} \sum_{\mathbf{r}} \left[\left\| (1 - \mathbf{A}(f'_B, R_2^{*'}) \mathbf{A}^+(f'_B, R_2^{*'})) \mathbf{s} \right\|_2^2 + \sum_{\mathbf{r}' \in \mathcal{N}(\mathbf{r})} U(|f'_B(\mathbf{r}) - f'_B(\mathbf{r}')|^2) \right], \quad (14.9)$$

where the least-squares terms for all voxels are summed up and an “interaction term” U between each voxel (at discrete location $\mathbf{r}$) and its neighborhood voxels at $\mathbf{r}' \in \mathcal{N}(\mathbf{r})$ is added. Global minimization

problems like Eq. (14.9) can successfully be solved with graph-cut algorithms [16], first demonstrated for WFI in [17] and further developed and employed in specific applications in [18–20]. We will be describing next the most important methods employed for estimating the field-map in water–fat separation.

Assuming a simple single-fat-peak water–fat voxel signal model, the complex signal at the n -th echo is

$$\mathbf{s}_{\text{model}}(t_n) = \left(\rho_W + \rho_F e^{i2\pi \Delta f_p t_n} \right) e^{i2\pi f_B t_n}, \quad (14.10)$$

with $t_1, t_2, \dots, t_N$ the different echo times, Δf_p the chemical shift of fat, ρ_W , and ρ_F the complex signal of the water and fat components. To estimate the static field inhomogeneity f_B in Eq. (14.10), Glover et al. proposed a method where three echoes are recorded with phase shifts of $-\pi$, 0 , and π between the fat and water resonances, simplifying the maximum likelihood estimation to a closed-form problem [21]. However, this choice of phase shifts restricts the echo times to be fixed. The fixed echo times are accompanied by limitations such as the restriction in the selection of other imaging parameter, e.g., the size of the FOV in a multi-echo acquisition or increased total scan time. Furthermore, it has been shown that these phase shifts are not the optimal candidates to reach the maximum possible noise performance [22,12]. Additionally, the signal model in Eq. (14.10) accounts for one chemical shift specimen only and does not account for R_2^* -decay effects. These limitations can significantly reduce the quantitative performance of the parameter estimation [10,23].

Therefore, there has been considerable interest in alternative choices of echo shifts, the inclusion of a multi-fat-peak model, and R_2^* [10,23]. In the case of arbitrary echo-time shifts, the estimation of the local frequency shift can not be decoupled from the estimation of water–fat contribution due to the nonlinearity of the frequency shift within the signal model. To estimate the local frequency in case of arbitrary echo-time shifts and a nonlinear maximum-likelihood estimate, Reeder et al. introduced a method for iterative decomposition of water and fat with echo asymmetry and least-squares estimates (IDEAL, [22]). The IDEAL method consists of repeated linearizations of the original nonlinear problem, alternately estimating the water–fat signals and the field-map using a gradient descent scheme. However, a gradient descent-based method with a simple initialization of $f_B = 0$ in all voxels has problems dealing with large-field inhomogeneity. The problem with large-field inhomogeneity arises since the maximum-likelihood estimate is not only nonlinear but also nonconvex and potentially contains several local and global minima [24,25] as depicted in Fig. 14.3. The assumption of a moderate field inhomogeneity is not generally true (especially at higher field strengths) and can lead to water–fat swaps associated with a significant error in the field-map estimation as depicted in the first row of Fig. 14.5.

Several methods have been proposed to address the problem with large-field inhomogeneities. Yu et al. [24] proposed an extension to the original gradient descent-based method, where a slowly varying field-map is imposed by a region-growing scheme. Lu et al. [25] proposed a coarse-to-fine grid approach in addition to the region-growing algorithm to further improve robustness, and Hernando et al. [26] proposed a method using variable projection (VARPRO) to resolve the large-field inhomogeneities by calculating the cost function Eq. (14.8) over its period length. The VARPRO approach reveals all local minima, and the extraction of the global minima is possible (see Fig. 14.3). The extraction of the global minimum from the VARPRO residual will subsequently be referred to as gmVARPRO. Field-map and water–fat separated images obtained by the gmVARPRO method are depicted in the second row of Fig. 14.5. However, all the aforementioned methods still can fail to correctly estimate the field-map in challenging anatomies or are prone to noise due to their voxel-independent fitting scheme.

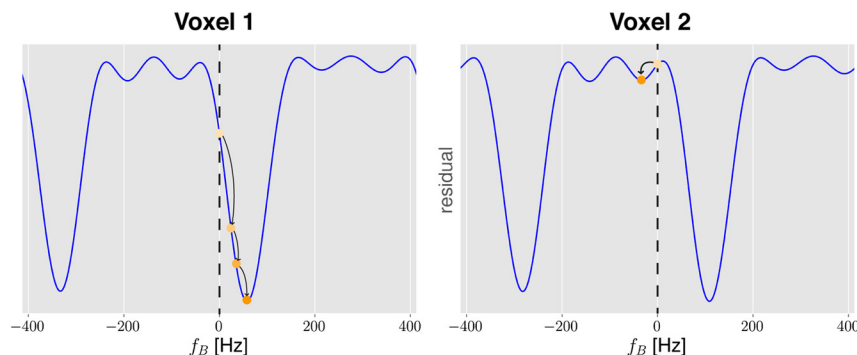


FIGURE 14.3 The voxel residual space of Eq. (14.8) of two voxels

The figure shows the VARPRO residual space of Eq. (14.8) in dependence of the field-map parameter f_B of two voxels from the knee data set shown in Fig. 14.5. Both voxel reveal the nonconvex (multiple minima) and periodic nature of the field-map estimate. In voxel 1, a gradient descent-based optimization is able to find the global minimum because the 0 initialization (dashed gray line) is within the correct valley. However, in voxel 2 the field-map shift is too large, and the 0 initialization is within a valley adjacent to the global minimum.

In the past decade, algorithms formulating the field-map estimation as a graph search have emerged and have proven to be particularly successful in solving the constrained optimization problem, using either min-cuts iteratively as proposed by Hernando et al. [17], single-min-cut approaches proposed by Cui et al. [20,27] and Boehm et al. [28], or a iterative fine to coarse-grid approach proposed by Berglund et al. [19]. These methods either add neighborhood regularization to enforce the field-map to be slowly varying [17,20,28] or use a threshold value for the maximal allowed field-map variation between adjacent voxels [27] or smoothing by a fine-to-coarse grid approach [19].

While significantly improving field-map estimation and water-fat separation, most proposed graph-cut methods still have intrinsic limitations. The iterative graph cut in [17] does not necessarily converge to the global minimum of its defined cost function, smooths the field-map by construction, and allows for adopting only two-dimensional neighborhood information in the field-map smoothness-constraint term (inter-slice regularization is not possible). The iterative fine-to-coarse grid approach in [19] is restricted to compare only two candidates per voxel and can consequently miss the correct solution. The single-min-cut in [27] labeled as *rapid Globally Optimal Surface Estimation* (rGOOSE) tried to address these problems. The rGOOSE method: i) Does not smooth the field map since it restricts the field-map candidates to be only local minima of the voxel-wise field-map estimate; ii) it also allows for 3D neighborhood regularization; and iii) it theoretically yields the exact solution of its defined cost function. However, the rGOOSE method is associated with long computation times and can still sometimes miss the optimal solution due to limitations in its underlying graph construction. The most recently proposed method by Boehm et al. [28] based on a so-called variable-layer, single-min-cut graph cut: i) Reportedly overcomes the limitations in graph construction in the rGOOSE method; ii) uses a novel data consistency formulation in its global problem formulation; iii) shows excellent field-mapping and water-fat separation results in a plethora of imaging situations; and iv) yields wrap-free field-maps. The exemplary illustration of the graph construction based on VARPRO residuals for the

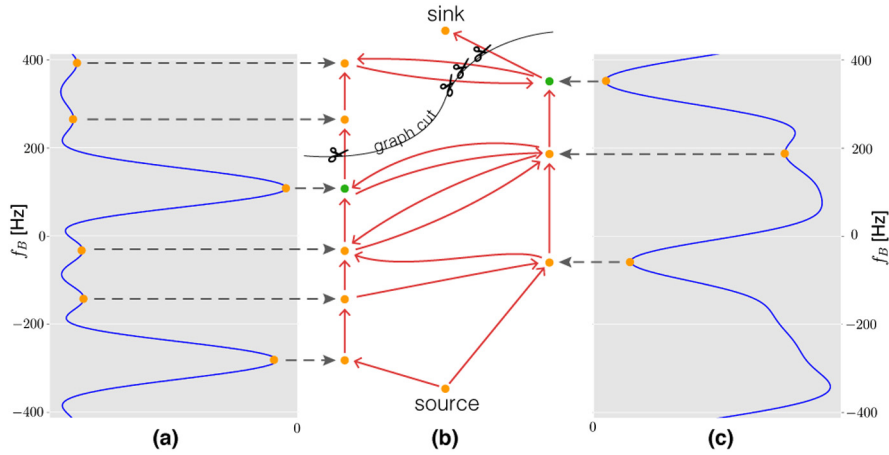


FIGURE 14.4 Exemplary illustration of the minima extraction and graph construction of two adjacent voxels in the graph cut based method field-mapping method by Boehm et al. [28]

The figure shows the residual space of two randomly selected adjacent voxels from the knee data set shown in Fig. 14.5. From the first voxel (a) 6 minima and from the second voxel (c) 3 minima are extracted, respectively. Based on the extracted minima a graph is constructed (b) that encodes the cost function (14.9) into its graph representation. Subsequently, the cost function is minimized by searching for the minimum cut in graph and the global optimal minima (green (light gray in print version) node) are identified. Please be note that the horizontal axis direction is flipped between (a) and (c).

method by Boehm et al. is shown in Fig. 14.4. Please note that the graph construction is always tailored to the specifics of the cost function to be minimized. Hence the graph for the different graph cut-based methods can be significantly different. The characteristic of the method by Boehm et al. to directly yield wrap-free field-maps is especially of high importance for field-map-based parameter estimation, such as quantitative susceptibility mapping (QSM) (Chapter 16), where wraps are expected to translate in strong susceptibility artifacts [29]. Field-map and water–fat separated images obtained by the state-of-the-art variable-layer single-min-cut graph cut method are depicted in the last row of Fig. 14.5.

14.3.3 Noise performance analysis

The second important consideration in solving (14.8) (or the first term in the global estimation (14.9)) is the noise propagation in the optimization, which depends on the echo-time selections in combination with the physical tissue parameters. At certain unfavorable combinations of echo times and model parameters, the noise in the input data s amplifies in the parameter estimates, which renders them useless for later diagnostic or analytic purposes. It is therefore important to choose well-suited echo times for the expected tissue parameters to optimally design the experiment before the data acquisition. The standard method for such an optimal experimental design in terms of echo-time selection and model parameters is the Cramér–Rao analysis [30,31], based on the computation of the Fisher information matrix (FIM) [32] for the chosen signal model. The FIM is defined as the expectation value ($\mathbb{E}[\dots]$) of

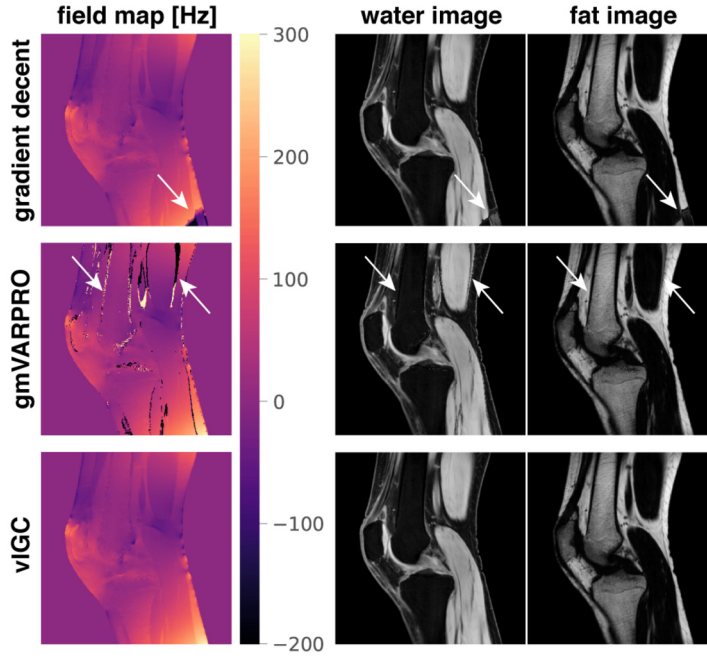


FIGURE 14.5 Field-map and water–fat separated images obtained by a voxel-wise gradient descent method [22], the gmVARPRO method [26] and a state-of-the-art variable layer graph cut (vIGC) [28]

The voxel-wise gradient-descent method (first row) is able to estimate field-map and water–fat separated images to a large extent correctly. However, at the bottom right corner of the FOV, the field-map variation is too large, and a field-map wrap and water–fat swap occurs in the corresponding maps. In contrast to the gradient descent method, the gmVARPRO method (second row) is able to resolve the large field-map variation at right left bottom of the FOV by extracting the global minimum of the residual space (compare with Fig. 14.3) per voxel. However, at regions with low signals such as the cortical bone shell, noise-like artifacts are visible in field-map indicating water–fat swaps. Furthermore, partial swaps occur in the whole FOV where water- or fat-only regions are falsely estimated to contain both water and fat. Field-mapping based on a vIGC method (last row) is able to correctly resolve the large field-map variation at the bottom of the FOV, has no noise-like estimates, and can correctly separate water and fat contributions throughout the whole volume.

the second derivative of the log-likelihood $\ln \mathcal{L}$

$$\mathbf{I}_{kl} = \mathbb{E} \left[\frac{\partial}{\partial \beta_l} \frac{\partial}{\partial \beta_k} \ln \mathcal{L} \right], \quad (14.11)$$

where the likelihood $\mathcal{L}$ in the case of additive white Gaussian noise with variance σ^2 follows

$$\mathcal{L} \sim \exp \left(-\frac{1}{2\sigma^2} \|s - \mathbf{A}\boldsymbol{\rho}\|_2^2 \right), \quad \boldsymbol{\rho} = [W, F]^\top. \quad (14.12)$$

In general, the FIM is then given by

$$\mathbf{I} = \frac{1}{\sigma^2} \Re\{\mathbf{J}^\dagger \mathbf{J}\}, \quad (14.13)$$

where $\mathbf{J}$ is the Jacobian of the chosen signal model [33]. In the case of the widely used single- R_2^* signal model (14.4), the Jacobian that defines the FIM by (14.13) is

$$\mathbf{J} = \left[\frac{\partial \hat{s}}{\partial |W|}, \frac{\partial \hat{s}}{\partial |F|}, \frac{\partial \hat{s}}{\partial \angle W}, \frac{\partial \hat{s}}{\partial \angle F}, \frac{\partial \hat{s}}{\partial f_B}, \frac{\partial \hat{s}}{\partial R_2^*} \right], \quad (14.14)$$

where $\angle W$ and $\angle F$ refers to the angle of W and F , respectively.

The Cramér–Rao lower bound (CRLB), which gives the theoretically minimal variances of the parameter estimates [33], is defined as

$$\text{CRLB} = \text{diag } \mathbf{I}^{-1} = [\text{Var}|W|, \text{Var}|F|, \text{Var}\angle W, \text{Var}\angle F, \text{Var}f_B, \text{Var}R_2^*]^\top. \quad (14.15)$$

Minimizing the CRLB for the model parameters of interest and/or different echo samplings can be used for optimal design [34] of the experimentally selected echo times. Again, the CRLB's in (14.15) are specific for the single- R_2^* water–fat signal model (14.4), but can be generalized for the class of summed cisoid (i.e., complex exponential) signal models [12].

The noise efficiency can also be quantified by the effective number of signals averages (NSA) defined as the variance in each of the measured images over the variance of a given parameter [34]. See Fig. 14.6 for an example.

14.4 Water–fat separation in non-Cartesian imaging

Non-Cartesian trajectories such as radial or spiral are more robust towards motion artifacts compared to Cartesian trajectories due to their dispersed distribution of motion artifacts [35–37]. For example, radial sampling is able to eliminate k-space gaps that are caused by motion-induced phase shifts, by repeated sampling of the k-space center. However, substantial motion still remains a challenge for non-Cartesian imaging and can result in blurring and aliasing artifacts, which appear as streaks for example when using radial trajectories [38,39]. An advantage of many non-Cartesian acquisitions, such as radial Stack-of-Stars (SoS) or spirals, is the continuous passage of the radial lines through the k-space center, allowing for retrospective self-gating and thus eliminating the need to use navigator signals or external devices for motion correction [40,41]. Especially, the use of golden-angle-ordered 3D SoS trajectories has been investigated for free-breathing water–fat imaging [42–44].

While being less sensitive to motion artifacts, non-Cartesian trajectories are less robust to gradient errors and delays than Cartesian ones (see Chapter 4). Different non-Cartesian trajectories, such as radial [42,45,46,43], spiral [47–49], PROPELLER [50,51], and concentric rings [52], have been investigated for water–fat separation. Besides gradient errors and delays, off-resonance effects affect the use of non-Cartesian trajectories for water–fat imaging. The origination of the blurring of the fat signal and its implications for non-Cartesian trajectories will be explained briefly in this section.

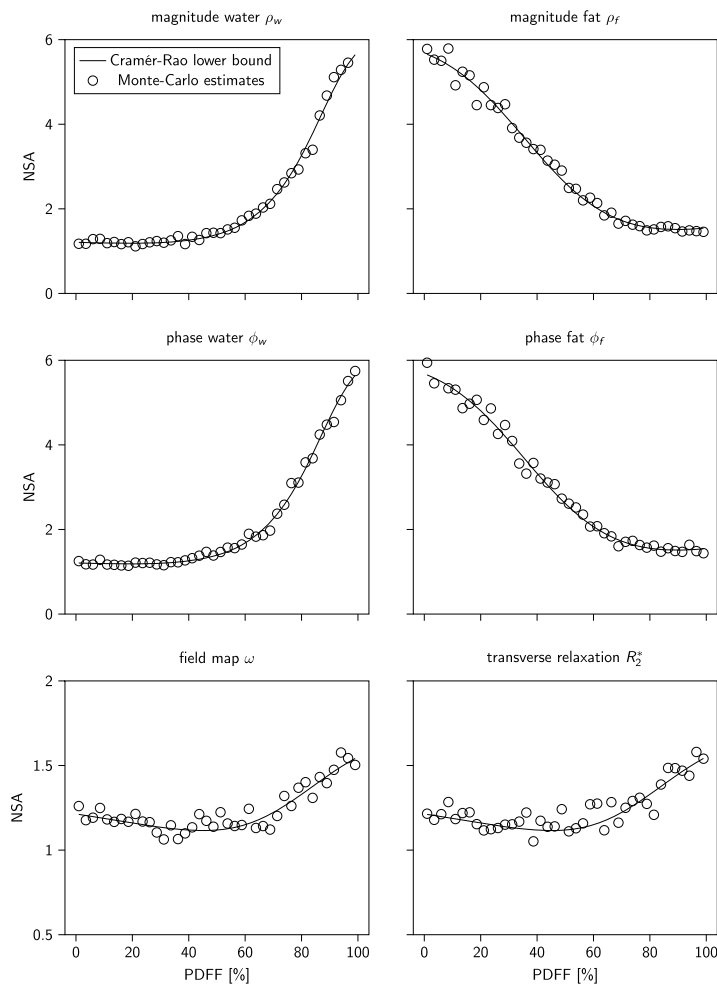


FIGURE 14.6 WFI simulation

Cramér–Rao lower bounds with Monte-Carlo estimates of parameter noise in a single- R_2^* water–fat model with a multi-peak fat model. Experimental setting is defined by: number of echoes $N_{TE} = 6$, first echo time $TE_1 = 1$ ms, echo spacing $\Delta TE = 1.2$ ms, $B_0 = 3$ T, $R_2^* = 5$ s $^{-1}$, and field-map $\omega/2\pi = 10$ Hz. The Monte-Carlo simulation was performed with 1000 independent noise realizations with signal-to-noise-ratio (SNR) of 100.

14.4.1 Water–fat shift artifact

The different Larmor frequencies of water and fat cause a misregistration of fatty tissue along the frequency-encoding direction (read or slice selection).

As discussed in Section 14.2, the signal from lipid protons will have a lower frequency compared to that of water protons. When the resonance frequency of the system is set to water, the signal from the lipid protons will appear to originate from water protons from a voxel from a lower part of the field.

The difference between the precession frequencies $\Delta f_{fw}(x)$ is given by

$$\Delta f_{fw}(x) = f_f - f_w = -\sigma_{fw} \frac{\gamma}{2\pi} B_0, \quad (14.16)$$

where the suffixes w and f stand for water and fat respectively σ_{fw} is the chemical shift between water and fat as a fraction of the field B_0 and γ is the gyromagnetic ratio [53]. If the bandwidth per voxel Δf_{voxel} is not much greater than $\Delta f_{fw}(x)$, this shift of the fat signal becomes visible in the resulting image. The fat signal is then spatially misregistered and invades into neighbored voxels in the read-and-slice selection directions. The bandwidth per voxel is defined as

$$\Delta f_{voxel} = \frac{\gamma}{2\pi} G_R \Delta x, \quad (14.17)$$

with Δx being the voxel size and G_R the readout gradient strength. The spatial shift of fatty tissue can then be calculated with

$$\Delta x_{shift} = \frac{\Delta f_{fw}}{\Delta f_{voxel}} \Delta x = N_{shift} \Delta x. \quad (14.18)$$

As long as the readout gradient is sufficiently large so that $N_{shift} < 0.5$, most of the fat signal remains within the voxel. Eq. (14.18) illustrates that, the lower the bandwidth per voxel, the larger the spatial misregistration artifact, and thus the fat-shift artifact.

For conventional Cartesian spin-warp imaging, the bulk shift of fat due to the chemical shift artifacts is well-understood and to some degree clinically acceptable. Specifically, in conventional Cartesian spin-warp imaging, the phase accumulation is identical in each echo and varies linearly across k-space in the frequency-encoding direction. However, in non-Cartesian imaging, the phase accumulated by off-resonant spins leads to distortion and highly degraded image quality. The blurring from the off-resonance fat signal has long challenged the clinical adoption of non-Cartesian acquisitions, especially in protocols employing low bandwidths and long readouts.

14.4.2 Fat blurring in non-Cartesian acquisitions

For most non-Cartesian acquisitions, the frequency-encoding direction varies during the acquisition and can cover all possible directions in k-space. While for most Cartesian trajectories the frequency-encoding dimension is fixed to one k-space direction, it changes, e.g., step-wise with every spoke in a radial stack-of-stars trajectory or even continuously for a spiral trajectory. Consequently the off-resonant fat signal is not bulk shifted any more in one direction in image space with regards to the water signal as in the Cartesian case, but instead the fat signal is shifted in all frequency-encoding directions that are used for the non-Cartesian trajectory (Fig. 14.7). This leads to blurring and distortions of the fat signal when using non-Cartesian trajectories.

The blurred fat signal can obscure underlying or neighboring pathologies including tumors, inflammation, and edema [46]. Especially, the longer readouts that are needed for spiral trajectories and low bandwidths are heavily affected by blurring and distortions of the fat signal since the longer the readout, the more off-resonant phase is accumulated. Consequently, short readouts or fat suppression are

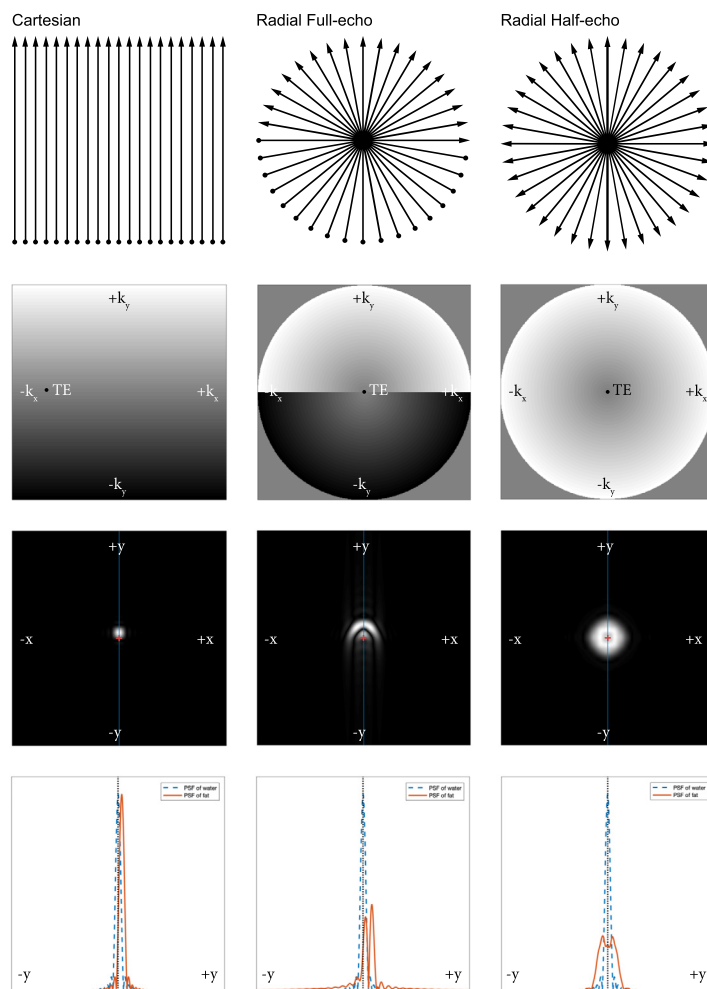


FIGURE 14.7 Simulation of the point spread function (PSF) of the fat signal for various sampling trajectories

Exact knowledge of the acquisition time for each k-space point is necessary to remove phase errors due to the off-resonance of fat. Graphical acquisition time maps in k-space are shown in the second row under the corresponding trajectories. Dark values represent a negative time shift respectively to TE, which means that those k-space points were sampled earlier than the actual TE time, while brighter values were sampled later than TE. The third row shows the PSF for the off-resonant fat components. For the Cartesian case, the PSF is bulk shifted in the frequency-encoding direction, while it is spread out for the two radial trajectories. Note that, for the unsymmetric full-radial case, the PSF also shows a clear directional dependence. 1D profiles of the PSF plotted along the blue (dark gray in print version) line in y-dimension are shown in the bottom row. The PSF of the water signal is added as a reference (blue (dark gray in print version) dotted line). Adapted from [46].

often needed for non-Cartesian imaging techniques to reduce the confounding blurred fat. On the other hand, fat suppression using inversion or spectrally selective pulses [54] can reduce the signal-to-noise ratio (SNR) or perform poorly in regions with B_0 inhomogeneities. To increase SNR, lower receiver bandwidths may be used that increase the water–fat shift and thus the blurring of the fat signal. Chemical shift encoding-based water–fat separation methods can be effective in removing the fat blurring in non-Cartesian acquisitions.

14.4.3 k-space-based water–fat separation

A direct approach to correct for the off-resonance of the various fat spectrum peaks is therefore to already include the phase accumulation in the k-space signal model. Historically IDEAL [22]-based methods and other image-based separation methods are performed in the image domain and treat each echo as if it was sampled instantaneously at the echo-time point t_n at which the k-space center is sampled. But, since the readout time is a finite process, it can be added to the signal model in k-space for a more accurate water–fat separation. This is done by extending t_n by a second time parameter $\tau_{\mathbf{k},n}$, which represents the relative time between the acquisition of sample point $\mathbf{k}$ and the center of k-space [55]. Brodsky et al. [46] assume a signal model that ignores the effects of T_2^* decay and changes in the field-map variation between echoes. In this case, the B_0 can be demodulated from the signal, and it can be expressed in k-space as

$$s_n(\tau_{\mathbf{k},n}, \mathbf{k}) = \sum_{m=1}^M \rho_m(\mathbf{k}) e^{i2\pi \Delta f m (t_n - \tau_{\mathbf{k},n})}. \quad (14.19)$$

$\tau_{\mathbf{k},n}$ models the finite readout and specifies the exact sampling time point $t_n - \tau_{\mathbf{k},n}$ of each k-space point $\mathbf{k}$. Similar to the VARPRO approach explained in Section 14.3 in image space, the signal can then be decomposed into water and fat maps in k-space directly. The reconstructed fat maps from this water–fat separation do not show the typical non-Cartesian fat blurring due to off-resonance of the fat spectrum peaks [46].

A more analytical but therefore computationally demanding approach to correct for the water–fat shift using the same signal model idea is presented by Benkert et al. [42]. Water (W), fat (F), and B_0 field-map (Φ) are directly fitted to the radial k-space data y in one single step by solving the minimization problem for each coil element c and echo time t

$$\arg \min_{c,t} \|\mathbf{E}(W, F, \Phi)_{c,t} - y_{c,t}\|_2^2 + \lambda_W \|\mathbf{S}(W)\|_1 + \lambda_F \|\mathbf{S}(F)\|_1. \quad (14.20)$$

With $\mathbf{S}$ being the sparsifying transform, the last two terms are the compressed sensing [56] regularizer for the water and the fat signal. The desired parameters are transformed to k-space data via the forward operator $\mathbf{E}$

$$\mathbf{E}(W, F, \Phi)_{c,t} = \mathbf{FT}(C_c \cdot \exp(2\pi i \cdot \Phi \cdot t_n) \cdot W) + D(t) \cdot \mathbf{FT}(C_c \cdot \exp(2\pi i \cdot \Phi \cdot t_n) \cdot F). \quad (14.21)$$

The parameter $\tau_{k,n}$ is included here in the forward operator to model the various chemical shifts of the fat peaks and to account for the off-resonant blurring of fat due to the radial readout

$$D(t) = \sum_{m=1}^M \alpha_m \cdot e^{2\pi i \cdot \Delta f_m \cdot (t_n + \tau_{k,n})}. \quad (14.22)$$

Both presented methods were primarily used to perform water–fat separation for non-Cartesian trajectories. However, the modification of the k-space signal by modeling the off-resonances of the various fat model peaks makes it, in general, possible to correct for chemical shift artifacts in any acquisition. Finally, while the deblurring of fat may be more convenient in k-space, depending on the employed model, the water–fat separation may be performed in image space or in k-space.

14.5 Confounding factors in quantitative water–fat imaging

Quantitative water–fat separation requires the consideration of potential confounding factors, which are traditionally characterized by either their origin as physical effects and hardware characteristics or by their appearance as magnitude and phase errors [57–61].

Physical effects typically include B_0 inhomogeneities [21,26], T_2^* decay [62,23], T_1^* relaxation differences between the water and fat [63,64], spectral complexity of the fat model [23,65], susceptibility-induced resonance shifts [66], temperature-induced resonance shifts of water [67], noise bias [63,64], and concomitant gradient effects [68,69]. Hardware imperfections include gradient delays [68] and non-flat frequency response characteristics of the receiver coils [58].

Gradient delays and concomitant gradients are two confounders that can be corrected during the reconstruction process and are described in the following section for Cartesian imaging trajectories.

14.5.1 Correction of hardware imperfections: gradient delays

In multi-echo gradient–echo-based water–fat imaging, multiple subsequent echoes are acquired in a train of echoes at different echo times. Due to hardware limitations and imperfections, the actual played out gradient waveform may deviate from the nominal defined gradient waveform. Consequently, this may lead to echo misalignments (and also higher-order effects which are not discussed further here), which may also differ across the echo train. For Cartesian trajectories, these echo misalignments often appear as an additional linear phase in the image domain along the readout direction (Fig. 14.8) according to the Fourier shift theorem

$$s(k - k_0) \longleftrightarrow S(r)e^{-i2\pi r k_0}. \quad (14.23)$$

This type of gradient delay can be estimated under the assumption that delays are symmetric with respect to the readout gradient polarity. Therefore, a reference measurement can be acquired by flipping the readout direction that corresponds to inverting the gradient polarity. Usually, it is sufficient to estimate the delay for the center k-space line only and to assume that the delay is independent of the phase-encoding gradients.

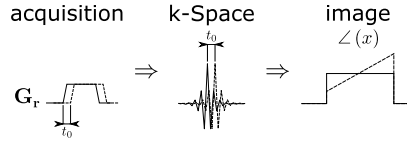


FIGURE 14.8 Effect of gradient delay in Cartesian imaging: k-space shifts propagate as a linear phase term into the image space according to the Fourier theorem

A gradient delay t_0 of the readout gradient G_r leads to an echo shift k_0 in k-space. According to the Fourier shift theorem, an echo shift k_0 in k-space leads to a linear phase along the readout direction in image space.

Therefore, the echo misalignment between opposite readout polarities can be estimated in k-space and corrected using the following optimization problem [68]:

$$k_0 = \arg \min_{k_0^*} \left\| \frac{\partial \left| \mathbf{F} \left(p^+ e^{(-i\pi k_0^* x)} \right) \right|}{\partial k} - \frac{\partial \left| \mathbf{F} \left(p^- e^{(+i\pi k_0^* x)} \right) \right|}{\partial k} \right\|_2, \quad (14.24)$$

where p^+ and p^- are the reference measurements (complex 1D image space profiles) acquired with opposite readout polarities and $\mathbf{F}$ is the Fourier transform. In some cases, the precision of the estimation can be further improved by zero padding the reference measurements in image space before the Fourier transform is applied.

The corresponding linear phase $\Delta \Phi k_0^n$ that is a function of x in image space and the n th echo can be computed as

$$\Delta \Phi k_0^n(x) = -\pi k_0^n x \quad (14.25)$$

and demodulated from the complex images.

Noticeable gradient delays typically occur when the immediate gradient waveform history of relevance for short-time eddy currents varies when comparing echoes. Typically, this can be the case when the first echo of a multi-echo gradient-echo train is compared with subsequent echoes of the train. An example of the effect on water–fat parameter estimation is given in Fig. 14.9.

14.5.2 Correction of concomitant gradients

According to Gauss’s law of magnetism, a magnetic-field gradient played out along a spatial axis is accompanied by concomitant gradients in the other two axes. Historically, the concomitant gradient field was first described as a confounding factor at low field strengths [70,71] and was then also described at higher field strengths in the context of phase contrast angiography [72], echo planar imaging [73,74] fast spin-echo imaging [75], balanced steady-state free-precession imaging [76], spiral imaging [77], and diffusion tensor imaging [78]. The correction of the concomitant gradient field-induced phase accumulation $\Delta \Phi_C$ was also investigated in CSI evaluating the influence on the estimation of PDFF [68,69], T_2^* [69,79] and B_0 mapping [69]. The effect of the concomitant gradient field on the water–fat signal model differs depending on the employed acquisition scheme: the additional phase term adds up

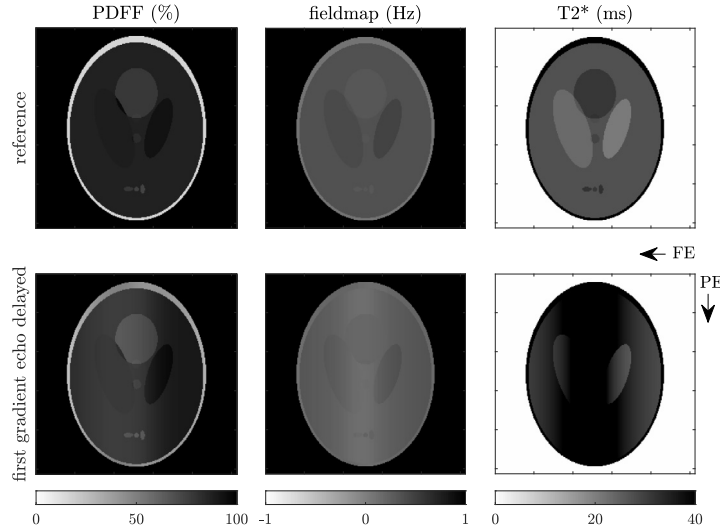


FIGURE 14.9 Exemplary parameter bias caused by a gradient delay of the first (out of six) gradient echo in Cartesian imaging

The upper row shows the reference parameter maps for PDFF, field-map, and T_2^* for a numerical phantom. The lower row shows the introduced bias (when compared to the upper row) by an echo delay of one k-space sample. Simulations were performed for a six-echo gradient-echo readout with a delta TE of 1 ms and first TE of 1 ms. This effect can be observed due to short-time eddy-current effects that are differently affecting the first gradient-echo compared to subsequent gradient-echoes in a multi-echo gradient-echo train. Frequency (FE) and phase encoding (PE) directions are indicated by black arrows.

to the field-map term for single TR measurements (where all echoes are acquired in a single TR) and causes a PDFF bias for interleaved acquisition schemes (where echoes are acquired over multiple TRs).

The phase term $\Delta\Phi_C$ can be approximated by an expression introduced by Bernstein et al. [72]

$$\Delta\Phi_C(x, y, z, t_n) = A(t_n)z^2 + B(t_n)(x^2 + y^2) + C(t_n)xz + D(t_n)yz, \quad (14.26)$$

where $A(t_n)$, $B(t_n)$, $C(t_n)$, and $D(t_n)$ are defined as

$$A(t_n) = \frac{\gamma}{2B_0} \int_0^{t_n} G_x(t)^2 + G_y(t)^2 dt \quad (14.27a)$$

$$B(t_n) = \frac{\gamma}{8B_0} \int_0^{t_n} G_z(t)^2 dt \quad (14.27b)$$

$$C(t_n) = -\frac{\gamma}{2B_0} \int_0^{t_n} G_x(t)G_z(t) dt \quad (14.27c)$$

$$D(t_n) = -\frac{\gamma}{2B_0} \int_0^{t_n} G_y(t)G_z(t) dt, \quad (14.27d)$$

where t is the time after excitation, γ is the gyromagnetic ratio, B_0 is the main magnetic field, and $G_x(t)$, $G_y(t)$, $G_z(t)$ denote the played out gradient waveforms in the physical coordinate system of the gradient system. As can be depicted from Eq. (14.27), the coefficients A and B arise from gradients G_x and G_y and G_z , respectively, whereas C and D are cross terms of the G_x and G_y with G_z , respectively. Therefore, the more gradient axes are used, the more complex the phase term $\Delta\Phi_C$ becomes.

Simplified examples of concomitant gradient-induced spatially dependent dephasing $\Delta\Phi_C$ for multi-echo gradient-echo readout in X , XZ and XYZ direction are given in Fig. 14.10, 14.11 and 14.12, respectively.

14.5.3 Proton density fat-fraction determination

After an optimal design of the experimental scenario based on CRLB analysis, an adequate field-map initialization and consideration of confounding factors, clinically important biomarkers can then be determined, e.g. tissue R_2^* or the proton density-fat fraction (PDFF). The PDFF can be estimated using either magnitude-valued or complex-valued signal-based techniques. While magnitude-based techniques are insensitive to phase errors (including gradient delays and concomitant gradients effects), complex-based methods are favored due to their superior noise performance.

The PDFF is typically computed by the magnitude-discrimination method [80] for complex-valued signal-based techniques as

$$\text{PDFF} = \begin{cases} 1 - \frac{|W|}{|W+F|} & \text{for } |W| > |F| \\ \frac{|F|}{|W+F|} & \text{for } |W| \leq |F|, \end{cases} \quad (14.28)$$

which is generally more noise robust against phase errors in the input images than simply computing the PDFF as $|F|/(|W| + |F|)$.

14.6 Current challenges and future directions

Despite the large growth of water–fat separation techniques over the last two decades, some exciting future directions exist for combining the method with many of the concepts emerging especially within the arena of quantitative imaging.

First, efficient fat suppression remains a challenge in acquisitions employing non-steady-state pulse sequences and requiring simulation of spin-history effects. The adoption of chemical shift encoding-based water–fat separation in such acquisitions is an emerging field. Techniques that employ nonsteady-state sequences and rely on undersampled non-Cartesian acquisitions, like Magnetic Resonance Fingerprinting (MRF), can be prone to fat blurring and can particularly benefit from Dixon imaging in achieving efficient fat suppression in body applications [81,82].

Second, water–fat separation has also been increasingly applied in quantitative imaging of multiple body tissues. Specifically, a quantitative imaging approach that aims to quantify the properties of the water component requires the application of efficient water–fat separation approaches to assure complete suppression of the fat signal [83–85].

Third, quantitative imaging of lipids and adipose tissue remains a major area of application of water–fat separation subject to growing interest. After the consideration of confounding effects, fat

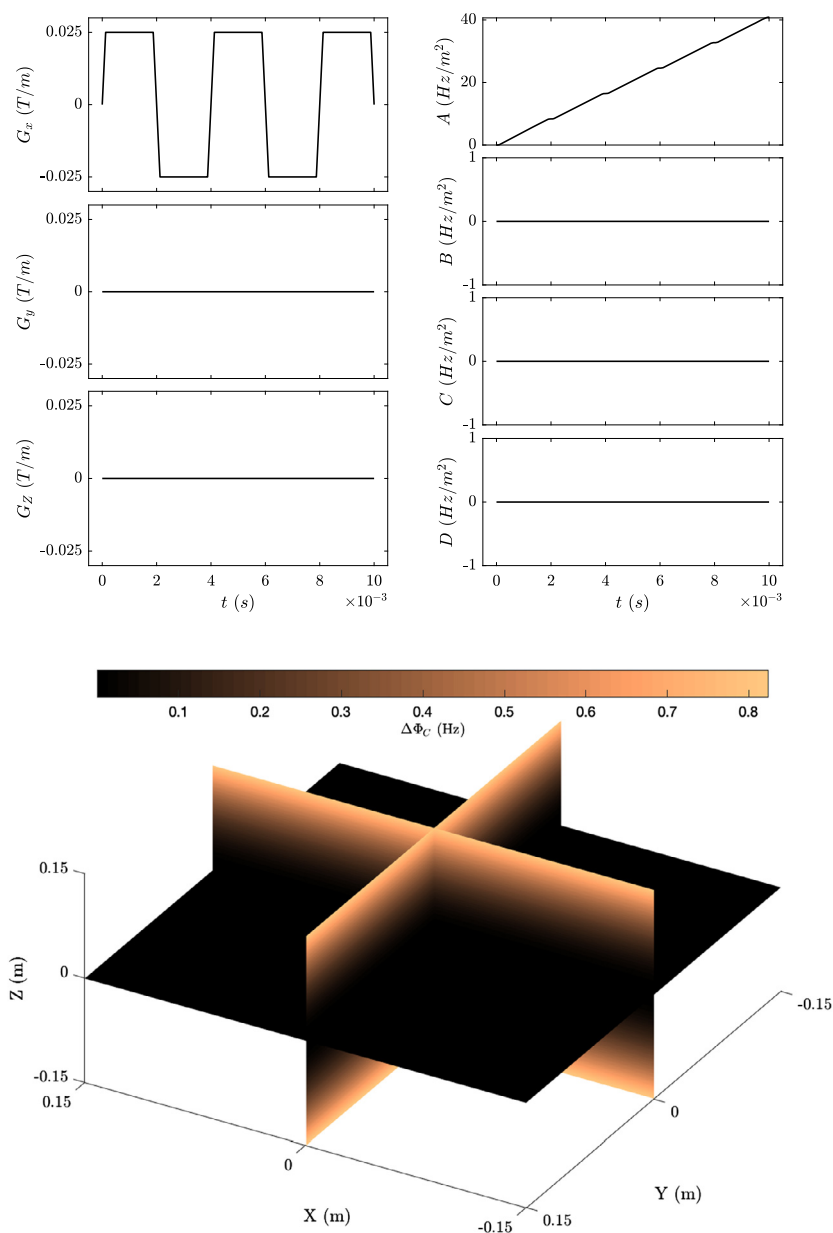


FIGURE 14.10 Spatial dependent dephasing due to concomitant gradients for readout gradients played out in X

Gradient waveforms G_x , G_y and G_z (represented in the scanner coordinate system XYZ), derived concomitant gradient coefficients A , B , C and D and resulting spatially dependent dephasing $\Delta\Phi_C$. The concomitant gradient coefficients A , B , C and D , and $\Delta\Phi_C$ were calculated assuming $B_0 = 3$ Tesla. $\Delta\Phi_C$ is displayed for the 5th echo.

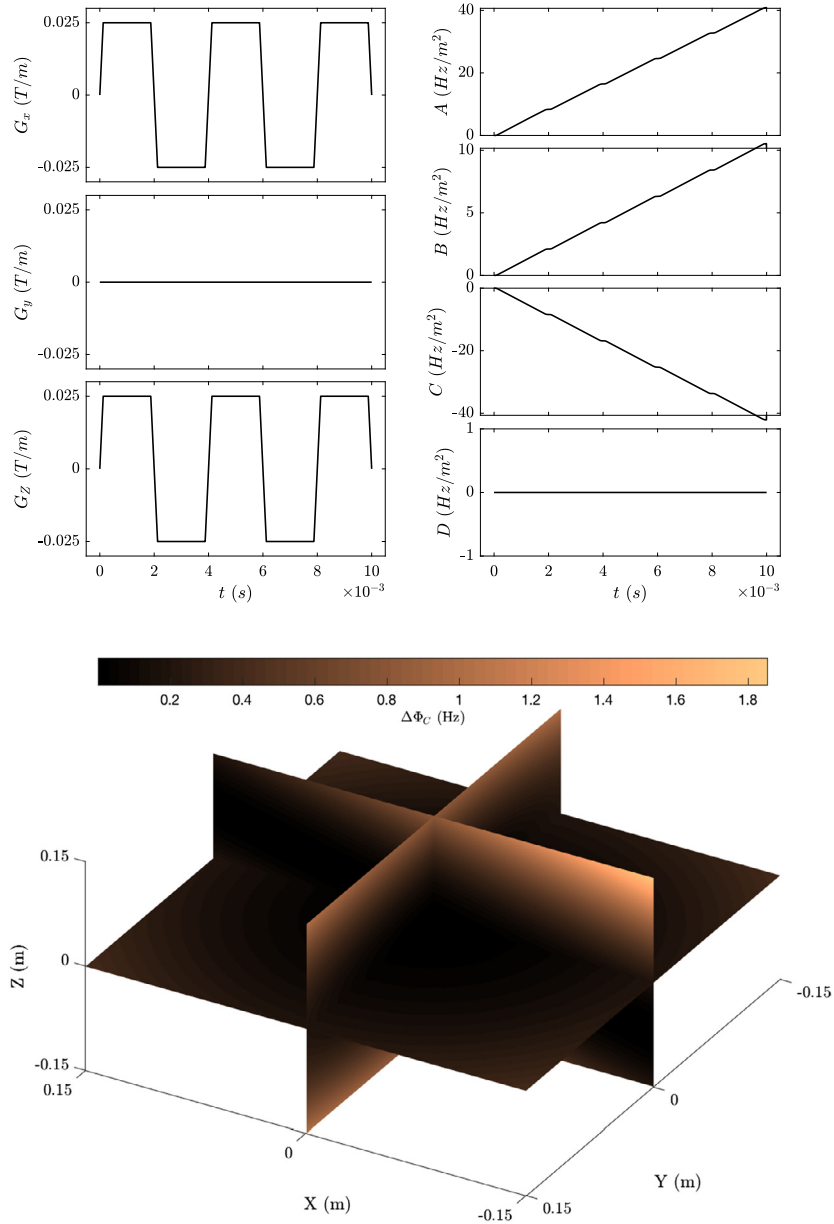


FIGURE 14.11 Spatially dependent dephasing due to concomitant gradients for readout gradients played out in XZ

Gradient waveforms G_x , G_y , and G_z (represented in the scanner coordinate system XYZ), derived concomitant gradient coefficients A , B , C and D and resulting spatially dependent dephasing $\Delta\Phi_C$. The concomitant gradient coefficients A , B , C , D and $\Delta\Phi_C$ were calculated assuming $B_0 = 3$ Tesla. $\Delta\Phi_C$ is displayed for the 5th echo.

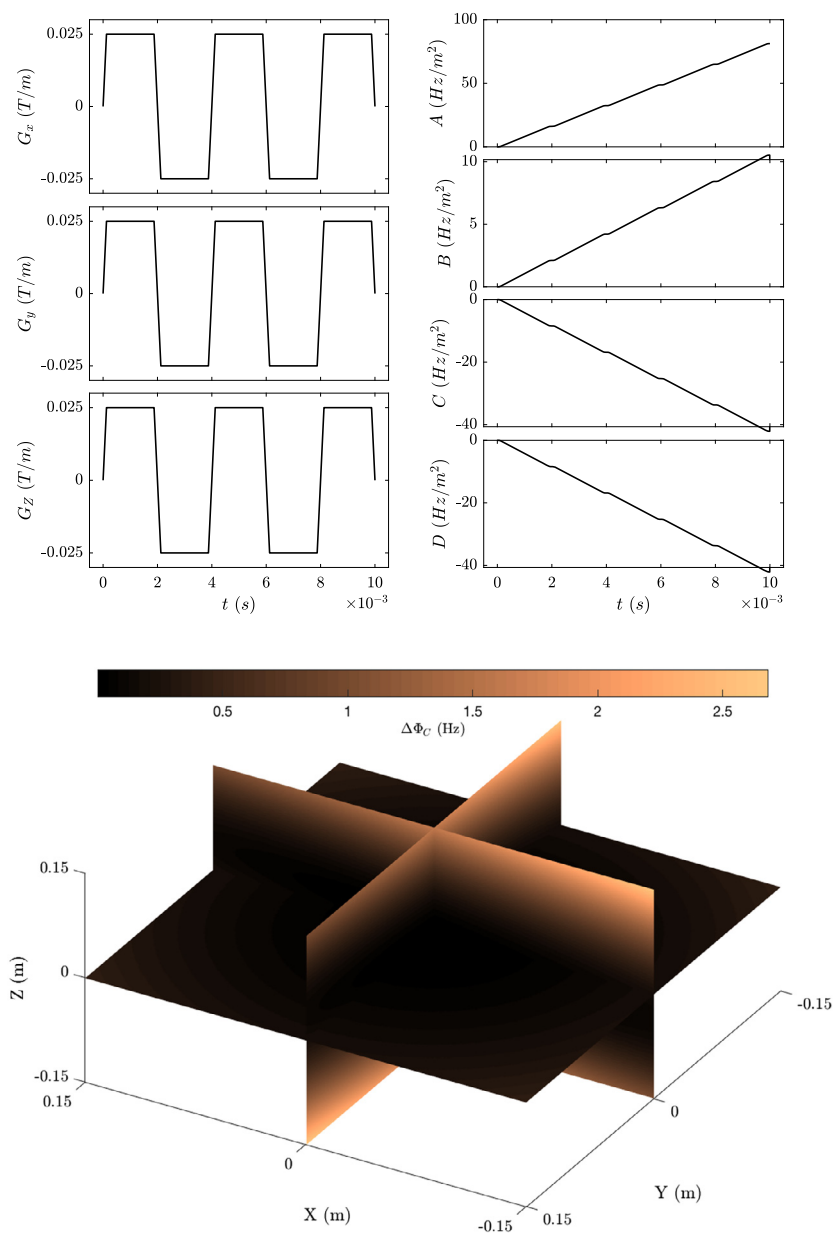


FIGURE 14.12 Spatially dependent dephasing due to concomitant gradients for readout gradients played out in XYZ

Gradient waveforms G_x , G_y , and G_z (represented in the scanner coordinate system XYZ), derived concomitant gradient coefficients A , B , C and D and resulting spatially dependent dephasing $\Delta\Phi_C$. The concomitant gradient coefficients A , B , C , D , and $\Delta\Phi_C$ were calculated assuming $B_0 = 3$ Tesla. $\Delta\Phi_C$ is displayed for the 5th echo.

quantification is nowadays a well-established technique for the liver [86]. However, a consensus on best practices and protocols for fat quantification beyond the liver is needed for the community [4].

14.7 Summary

Solving the water–fat separation problem requires primarily the treatment of a nonconvex optimization problem and the reliable estimation of the field-map. However, water–fat separation in general constitutes an image reconstruction problem that relies on a signal model that has to consider both the underlying data acquisition and spatial encoding process. A variety of signal models have been proposed for water–fat separation. Their complexity increases for the needs of PDFF quantification, as well as with the need for fat deblurring when applying water–separation in non-Cartesian acquisitions.

14.8 Further reading

In 2012, the International Society of Magnetic Resonance in Medicine (ISMRM) organized a scientific workshop on water–fat separation [87] where W. Thomas Dixon gave an enjoyable talk on how it all began. In addition, the workshop website is a good source to review some more detailed concepts and to get some hands-on practice with Matlab[®] code that was made available (ISMRM water–fat toolbox), together with example data.¹

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¹ <https://www.ismrn.org/workshops/FatWater12/data.htm>.

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